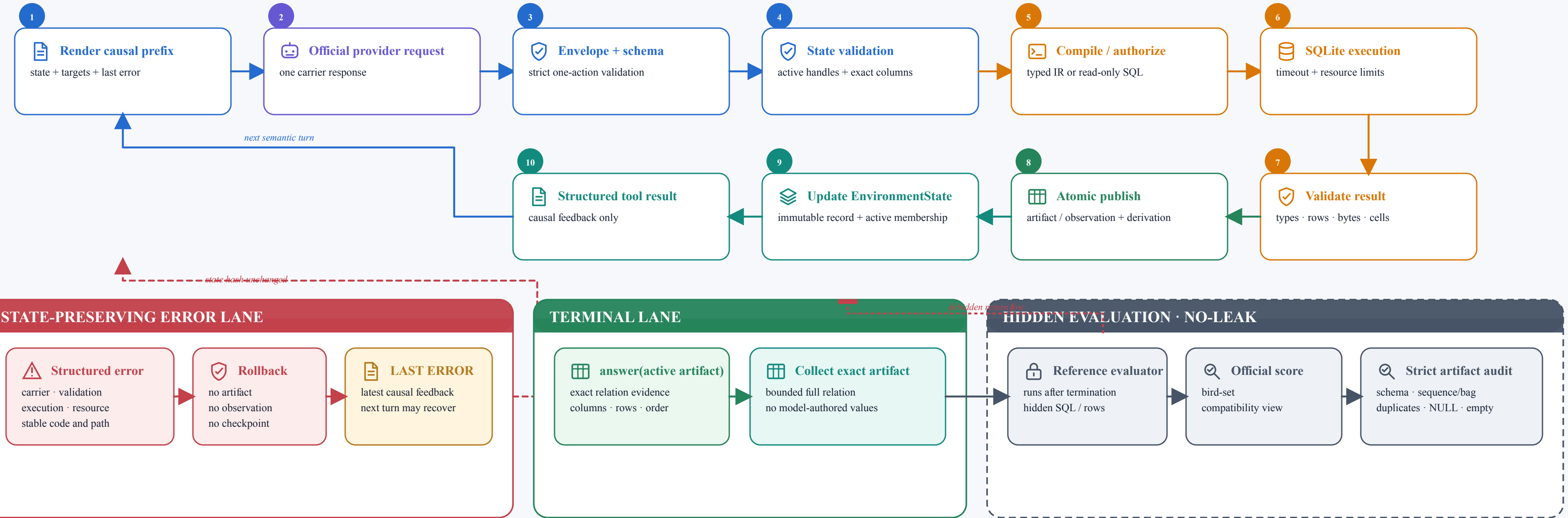


Causal Episode Loop, Safety Invariants, and Admission Boundary

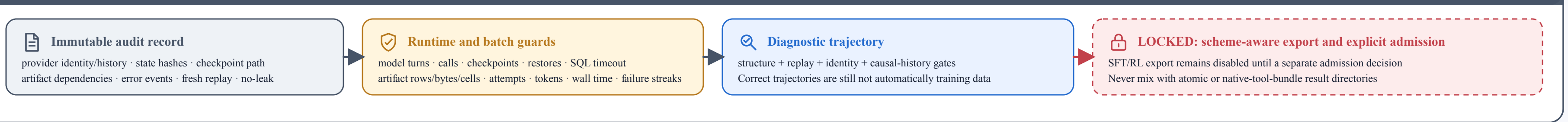
A real model↔Harness trajectory with state-preserving failures and isolated evaluation

DIAGNOSTIC ONLY

CAUSAL LOOP



AUDIT, RESOURCE GUARDS, AND ADMISSION



F10 · Success / failure state invariant

$$S_{t+1} = \begin{cases} \text{Apply}(S_t, a_t), & \text{valid}(a_t), \\ S_t, & \text{rejected}(a_t) \vee \text{failed}(a_t), \end{cases} \quad H(S_{t+1}) = H(S_t) \text{ on failure}$$

F12 · Independent strict artifact audit

$$\text{Strict}(\hat{R}, R^*) = \text{Schema} \wedge \text{Rows}_{\text{multiset/sequence}} \wedge \text{NULL} \wedge \text{Multiplicity}$$